AEGIS: Equilibrium Structure Mining for Certifiable Graph Robustness

Anonymous Author(s)

***Abstract*—We introduce AEGIS, a framework that mines the equilibrium structure of contractive implicit graph neural networks (IGNN-class models) to analyze adversarial vulnerability under graph structure perturbation. These models converge to a fixed point $z^* = F(z^*, A)$, enabling construction of a structural sensitivity matrix $S = (I - J_z)^{-1} J_A$ via the implicit function theorem. We derive a contraction-based vulnerability characterization around a critical budget $\varepsilon_{\text{crit}} = (1-\kappa)/\|W\|_2$, where $\kappa = \|J_z\|_2 < 1$ is the operator-norm contraction constant: below it, fixed-point shifts are bounded with tight first-order predictions; above it, the contraction certificate becomes void. Under realistic perturbation constraints (symmetric, edge-only), we construct a constrained sensitivity matrix S_c whose first-order predictions achieve tightness 1.00 ± 0.01 at $\varepsilon = 0.01$ across 9 datasets (5 graph benchmarks + 4 IEEE power grids, 10 seeds each). AEGIS's SVD-optimal attack inflicts 20–50% more damage than adaptive white-box PGD across all tested budgets. Its deterministic first-order per-node robustness radii provide structurally differentiated vulnerability assessments, unlike randomized smoothing which yields uniform bounds at comparable noise levels. As a case study, the vulnerability spectrum recovers power grid N-1 contingency rankings (precision@10 = 0.66–0.81). Code will be released upon publication.**

***Index Terms*—Deep equilibrium models, graph neural networks, adversarial robustness, structural sensitivity, graph perturbation, implicit function theorem**

I. Introduction

Graph neural networks (GNNs) are vulnerable to adversarial structural perturbation: small changes to the graph topology—adding or removing a few edges—can drastically alter predictions [1], [2]. Defending against such attacks requires understanding *which perturbations matter* and *by how much*. Current approaches either provide loose certified bounds via randomized smoothing [3] or rely on heuristic attacks without guarantees [2].

We observe that a specific family of GNNs—contractive implicit GNNs (IGNN-class [4])—possesses a structural property that enables a qualitatively different kind of analysis. These models iterate a contractive operator F_θ until convergence to a fixed point $z^* = F_\theta(z^*, A)$, rather than propagating through a fixed number of layers. This fixed-point structure, combined with the implicit function theorem (IFT), yields the *structural sensitivity matrix*

$$S = (I - J_z)^{-1} J_A, \tag{1}$$

where $J_z = \partial F/\partial z|_{z^*}$ and $J_A = \partial F/\partial \text{vec}(A)|_{z^*}$. The matrix S maps structural perturbations to fixed-point shifts: $\Delta z^* \approx S \cdot \text{vec}(\delta A)$.

The central finding is that, under realistic perturbation constraints (symmetric, edge-weight-only), a *constrained* variant S_c of this sensitivity matrix predicts adversarial vulnerability with tightness 1.00 ± 0.01 to first order at small perturbation budgets ($\varepsilon = 0.01$). This holds across 9 datasets spanning 4 domains and 10 random seeds. The prediction is first-order; for large perturbations ($\varepsilon > 0.1$), higher-order terms dominate and the approximation becomes conservative. AEGIS applies specifically to contractive implicit GNNs (IGNN-class); explicit architectures such as GAT, GIN, and GraphSAGE lack the fixed-point structure required for this analysis.

Contributions. We make three contributions:

1) **Contraction-based vulnerability characterization.** For contractive implicit GNNs (IGNN-class [4]), we formalize vulnerability regimes around a critical budget $\varepsilon_{\text{crit}} = (1-\kappa)/\|W\|_2$ using the operator-norm contraction constant $\kappa = \|J_z\|_2 < 1$. We make assumptions explicit (ReLU, spectral-norm constraint, linear head) and quantify non-normality effects (section III).
2) **Constrained structural sensitivity.** We construct a constrained sensitivity matrix S_c that restricts analysis to symmetric, edge-only perturbations—the physically meaningful threat model for graphs. Under these constraints, first-order predictions achieve tightness ≈ 1.00. From S_c we derive SVD-optimal structural attacks (20–50% stronger than adaptive white-box PGD) and deterministic first-order per-node robustness radii (sections IV and V).
3) **Cross-domain validation.** We validate across 9 datasets in 4 domains (citation, e-commerce, encyclopedia, power flow). Constrained first-order tightness ≈ 1.00 holds across all domains. As a case study, the vulnerability spectrum recovers power grid N-1 contingency rankings with precision@10 = 0.66–0.81 (section VI).

II. Background

A. Deep Equilibrium Graph Neural Networks

A DEQ-GNN [5] defines a weight-tied operator $F_\theta : \mathbb{R}^{N\times d} \times \mathbb{R}^{N\times N} \to \mathbb{R}^{N\times d}$ and iterates it to convergence:

$$Z_{k+1} = F_\theta(Z_k, A), \quad Z_0 = 0. \tag{2}$$

When F_θ is contractive ($\kappa = \|J_z\|_2 < 1$, where $J_z = \partial F/\partial z$), the Banach fixed-point theorem guarantees a unique equilibrium $z^* = F_\theta(z^*, A)$. The Implicit Graph Neural Network (IGNN) [4] instantiates this with $F(Z, A) = \sigma(\hat{A} Z W^\top + X_{\text{proj}})$, where $\hat{A} = D^{-1/2}(A + I)D^{-1/2}$ is the normalized adjacency and W is spectral-norm constrained to ensure $\kappa \le \left\|\hat{A}\right\|_2 \cdot \|W\|_2 < 1$.

B. Implicit Function Theorem for DEQ Sensitivity

At the fixed point, the IFT provides the sensitivity of z^* to any parameter p:

$$\frac{\partial z^*}{\partial p} = (I - J_z)^{-1} \frac{\partial F}{\partial p}\bigg|_{z^*}, \tag{3}$$

where $J_z = \partial F/\partial z|_{z^*}$ is the state Jacobian. When $\kappa = \|J_z\|_2 < 1$, the resolvent $(I - J_z)^{-1}$ exists and is bounded by $\left\|(I - J_z)^{-1}\right\|_2 \leq 1/(1-\kappa)$ via the Neumann series. This bound uses the operator norm κ rather than the spectral radius $\rho(J_z)$; the latter yields the weaker bound $1/(1-\rho)$ only for normal J_z. The pseudospectral index $\eta = \left\|(I - J_z)^{-1}\right\|_2 \cdot (1-\rho) \geq 1$ [6] measures the gap, with $\eta = 1$ for normal matrices. In practice, IGNN's $J_z = \mathrm{diag}(\sigma') \cdot (\hat{A} \otimes W)$ exhibits mild non-normality ($\eta = 1.02$–1.28 across our experiments). The IFT has been used for hyperparameter optimization in implicit layers [7] and general implicit differentiation [8], but not for adversarial robustness analysis of graph structure.

C. Adversarial Structural Perturbation

We consider an attacker who perturbs the adjacency matrix: $A' = A + \delta A$ with $\|\delta A\|_F \leq \varepsilon$. For undirected graphs, δA must be symmetric. In graph terms, this corresponds to modifying edge weights or adding/removing edges. Unlike input-feature perturbation (studied for DEQs in [9], [10]), structural perturbation changes the *operator* itself, not just its input—the fixed point depends on A through both J_z and J_A.

III. Adversarial Equilibrium Theory

We formalize how structural perturbations affect DEQ-GNN equilibria, yielding actionable thresholds for practitioners. The characterization leverages the Banach fixed-point theorem and IFT; the central novel object is the *constrained sensitivity matrix* S_c (below), which enables tight first-order predictions under realistic perturbation constraints.

Theorem 1 (Vulnerability Characterization for Contractive Implicit GNNs)**.** *Let $F_\theta(Z, A) = \sigma(AZW^\top + X_{\mathrm{proj}})$ be an IGNN-class operator satisfying:*

(A1) *σ is ReLU (or any 1-Lipschitz activation with $\sup|\sigma'| = 1$);*
(A2) *W is spectral-norm constrained: $\|W\|_2 \leq c$;*
(A3) *The state Jacobian satisfies $\|J_z\|_2 \leq \kappa < 1$ at the fixed point z^* (contraction in operator norm).*

Define the critical budget

$$\varepsilon_{\mathrm{crit}} = \frac{1-\kappa}{\|W\|_2}. \tag{4}$$

Then structural perturbations δA exhibit three regimes:
(a) Subcritical *($\|\delta A\|_F < \varepsilon_{\mathrm{crit}}$): The perturbed operator remains contractive with a unique fixed point satisfying*

$$\|\Delta z^*\|_F \leq \sigma_1(S) \cdot \|\delta A\|_F + O(\|\delta A\|_F^2), \tag{5}$$

where $S = (I - J_z)^{-1} J_A$ and $\sigma_1(S) \leq \|J_A\|_{\mathrm{op}} / (1-\kappa)$.
(b) Critical *($\|\delta A\|_F \to \varepsilon_{\mathrm{crit}}$): The resolvent norm $\left\|(I - J_z')^{-1}\right\|_2$ diverges as $\Omega(1/(\varepsilon_{\mathrm{crit}} - \|\delta A\|_F))$. Since $\kappa = \|J_z\|_2$ already accounts for non-normality (unlike the spectral radius $\rho(J_z)$, which can underestimate the resolvent norm by a factor $\eta \geq 1$ [6]), the κ-based threshold requires no normality assumption.*
(c) Supercritical *($\|\delta A\|_F > \varepsilon_{\mathrm{crit}}$): The sufficient contraction certificate becomes void ($\|J_z'\|_2$ may exceed 1). The perturbed operator may still be contractive depending on perturbation direction, but the Banach fixed-point theorem no longer guarantees uniqueness, and all first-order guarantees from part (a) cease to apply.*

Proof. **Part (a).** The true Jacobian of F w.r.t. $\mathrm{vec}(Z)$ is $J_z = \mathrm{diag}(\sigma') \cdot (A \otimes W)$, where $\sigma' \in \{0,1\}^{Nd}$ for ReLU (A1). Thus $\kappa = \|J_z\|_2 \leq \|A\|_2 \cdot \|W\|_2 \cdot \sup|\sigma'| = \|A\|_2 \cdot \|W\|_2$. By the IFT applied to $G(z, A) = z - F(z, A) = 0$ at (z^*, A):

$$\Delta z^* = (I - J_z)^{-1} J_A \cdot \mathrm{vec}(\delta A) + O(\|\delta A\|^2). \tag{6}$$

Taking operator norms: $\|\Delta z^*\|_F \leq \left\|(I - J_z)^{-1}\right\|_2 \cdot \|J_A\|_{\mathrm{op}} \cdot \|\delta A\|_F + O(\|\delta A\|_F^2)$. Since $\left\|(I - J_z)^{-1}\right\|_2 \leq 1/(1-\kappa)$ (Neumann series, convergent because $\kappa = \|J_z\|_2 < 1$), this gives (5).

For contractivity preservation under perturbation: the perturbed Jacobian satisfies $\|J_z'\|_2 \leq \|A + \delta A\|_2 \cdot \|W\|_2 \leq (\|A\|_2 + \|\delta A\|_F) \cdot \|W\|_2$ (using $\|\cdot\|_2 \leq \|\cdot\|_F$). The condition $\|\delta A\|_F < (1-\kappa)/\|W\|_2$ ensures $\|J_z'\|_2 < 1$.

Part (b). As $\|\delta A\|_F \to \varepsilon_{\mathrm{crit}}$, $\|J_z'\|_2 \to 1$, so $\left\|(I - J_z')^{-1}\right\|_2 \geq 1/(1 - \|J_z'\|_2) \to \infty$. Unlike the spectral-radius bound $1/(1-\rho)$, which requires normality, the κ-based bound holds for all J_z with $\kappa < 1$. In our experiments, the pseudospectral index $\eta = \left\|(I - J_z)^{-1}\right\|_2 \cdot (1-\rho)$ ranges from 1.02 to 1.28, confirming mild non-normality.

Part (c). When $\|J_z'\|_2 \geq 1$, the contraction mapping condition fails. The Banach fixed-point theorem no longer guarantees existence or uniqueness; the iteration may converge to a different equilibrium, oscillate, or diverge. □

Remark. This characterization is conservative in two senses: (1) $\varepsilon_{\mathrm{crit}}$ is a sufficient, not necessary, condition—the perturbed system may remain contractive beyond $\varepsilon_{\mathrm{crit}}$ depending on perturbation direction; (2) the unconstrained bound uses worst-case $\|\cdot\|_F$ relaxation. Under constrained perturbations, the first-order prediction achieves tightness ≈ 1.00 empirically (section V-A). We use the operator-norm contraction constant $\kappa = \|J_z\|_2$ rather than the spectral radius $\rho(J_z)$; for normal J_z these coincide, while for non-normal J_z the κ-based bound is more conservative but requires no normality assumption.

Proposition 2 (Optimal First-Order Structural Attack)**.** *The perturbation δA^* maximizing $\|\Delta z^*\|$ to first order subject to $\|\delta A\|_F \leq \varepsilon$ is $\delta A^* = \varepsilon \cdot \mathrm{reshape}(v_1, N \times N)$, where v_1 is the leading right singular vector of S. The maximum first-order shift is $\varepsilon \cdot \sigma_1(S)$.*

Threat model and constrained sensitivity. We perturb the normalized adjacency $\hat{A}$ directly, subject to $\left\|\delta\hat{A}\right\|_F \leq \varepsilon$, symmetry ($\delta\hat{A} = \delta\hat{A}^\top$), and edge-only support. This models continuous edge-weight perturbations; discrete edge additions that change topology and degree normalization are outside the formal guarantee. We construct the *constrained sensitivity matrix* $S_c \in \mathbb{R}^{D\times|E|}$ with column $S_{:,iN+j} + S_{:,jN+i}$ for each edge $(i,j) \in E$, reducing the perturbation space from N^2 to $|E|$ dimensions. The constrained-optimal attack uses $\sigma_1(S_c)$, yielding tightness ≈ 1.00 empirically (section V-A). This S_c construction is the central technical contribution: it transforms unconstrained sensitivity analysis into a tight, practical tool for realistic graph perturbations.

Proposition 3 (Per-Node Robust Radius)**.** *Assume a linear classification head $f(z) = Wz+b$ (as in standard IGNN). For node v with margin $m_v = f_{y_v}(z_v^*) - \max_{c\neq y_v} f_c(z_v^*) > 0$, the first-order robust radius is*

$$r_v = \frac{m_v}{\|\partial f/\partial z_v^*\|_2 \cdot \|S_v\|_2}, \tag{7}$$

where S_v denotes the block-rows of S for node v, and $\|\partial f/\partial z_v^*\|_2 = \|W_{y_v} - W_{c^*}\|_2$ for the runner-up class c^*. Any $\|\delta A\|_F < r_v$ preserves the classification of node v. For nonlinear heads (softmax + MLP), the global Lipschitz constant must be replaced by the local Lipschitz constant at z_v^*, yielding a potentially tighter but harder-to-compute radius.

These radii are deterministic (not probabilistic), unlike randomized smoothing [3] which holds with probability $1-\alpha$. They are first-order guarantees: locally tight for small ε but increasingly conservative as perturbation magnitude grows and higher-order terms dominate. Without bounding the second-order remainder $O(\|\delta A\|_F^2)$, they should be interpreted as local sensitivity radii rather than global robustness certificates.

IV. AEGIS Framework

AEGIS packages the theory into a Python API applicable to any contractive implicit GNN (IGNN-class architecture). Given an operator F with factorized A/W structure, fixed point z^*, and context (adjacency, features), the `IEMiner` class provides:

1) **Vulnerability analysis**: compute S, $\sigma_1(S_c)$, and the per-edge vulnerability spectrum $v_{ij} = \|S_{:,iN+j} + S_{:,jN+i}\|$.
2) **First-order robustness radii**: per-node r_v via theorem 3, critical budget $\varepsilon_{\text{crit}}$ via theorem 1.
3) **Contractivity verification**: contraction constant $\kappa = \|J_z\|_2$ via largest singular value, non-normality index $\eta = \|(I - J_z)^{-1}\|_2 \cdot (1-\rho)$.

Subgraph-based analysis. The framework operates on BFS ego-subgraphs (typically 50–200 nodes) extracted from the full graph. The model is trained on the full graph; only the adversarial analysis is local. This suffices because per-node radii and edge vulnerability are inherently local quantities—a node's first-order radius depends on its own margin and the sensitivity of its neighborhood, not distant parts of the graph.

Computational cost. The dominant cost is computing $J_z \in \mathbb{R}^{D\times D}$ and $J_A \in \mathbb{R}^{D\times N^2}$ via row-by-row backward passes, followed by a linear solve for S. Total cost is $O(D\cdot N^2 + D^3)$ where $D = N\cdot d$. On a single GPU, this ranges from 0.5s ($N = 20$) to 8.1s ($N = 200$); see section V-F.

TABLE I: Cross-domain adversarial analysis (mean ± std, 10 seeds). Constrained tightness ≈ 1.00 across all domains.

Dataset	ρ	Tight.	AtkAdv	$\varepsilon_{\text{crit}}$	Cert%
Cora	$.33_{\pm.14}$	$1.01_{\pm.01}$	$3.6_{\pm.6}$x	$.66_{\pm.15}$	$81_{\pm\ 9}$%
Citeseer	$.59_{\pm.02}$	$1.01_{\pm.01}$	$4.1_{\pm.5}$x	$.41_{\pm.02}$	$83_{\pm\ 4}$%
Pubmed	$.41_{\pm.11}$	$1.01_{\pm.01}$	$3.2_{\pm.4}$x	$.59_{\pm.11}$	$74_{\pm23}$%
Amazon	$.14_{\pm.02}$	$1.00_{\pm.00}$	$3.8_{\pm.2}$x	$.86_{\pm.02}$	$92_{\pm\ 4}$%
WikiCS	$.34_{\pm.04}$	$1.00_{\pm.01}$	$3.8_{\pm.4}$x	$.66_{\pm.04}$	$70_{\pm\ 3}$%

V. Experiments

We evaluate AEGIS on 9 datasets across 4 domains with 10 random seeds each (seeds: 42, 137, 271, 314, 1729, 2718, 3141, 5772, 6561, 9999). All graph benchmark experiments use IGNN [4] with hidden dimension 64, spectral-norm constrained weights, and early stopping on validation accuracy over 200 epochs. Power flow experiments use ContractiveGCN-PF (an IGNN variant for AC power flow).

Notation. All formal bounds (theorem 1) use the operator-norm contraction constant $\kappa = \|J_z\|_2$. Tables report the spectral radius $\rho(J_z)$ as a diagnostic quantity. Since non-normality is mild ($\eta = 1.02$–1.28, section V-I), we have $\kappa \approx \rho$ in practice; the $\varepsilon_{\text{crit}}$ values computed with ρ are at most 28% optimistic relative to the κ-based formal threshold.

Datasets. Graph benchmarks: Cora (2,708 nodes), Citeseer (3,327), Pubmed (19,717), Amazon Photo (7,650), WikiCS (11,701). Power flow: IEEE case14 (14 buses), case30 (30), case57 (57), case118 (118), generated via PandaPower with 2,000 samples each.

A. Cross-Domain Adversarial Analysis

table I reports the constrained tightness ratio (actual shift / predicted shift at $\varepsilon = 0.01$), attack advantage over random perturbation, and per-node coverage (fraction with positive first-order radius) across 50-node BFS ego-subgraphs.

Tightness is 1.00 ± 0.01 across all datasets, confirming that the constrained first-order prediction is essentially exact under symmetric edge perturbations at $\varepsilon = 0.01$. The attack advantage (3.2–4.1× over random) demonstrates that AEGIS identifies the most vulnerable perturbation directions. Coverage (Cert%) ranges from 70% (WikiCS) to 92% (Amazon Photo), reflecting nodes with positive classification margin.

B. Attack Comparison: AEGIS vs. Mettack

We compare AEGIS's IFT-based structural attack against Mettack [2] (Meta-Self variant with 2-layer GCN surrogate). Both methods rank edges for removal; damage is measured by IGNN fixed-point shift $\|\Delta z^*\|$ after reconvergence.

AEGIS inflicts 3–10× more damage than Mettack at every seed and budget level (30/30 wins at $k = 1,\ldots,5$). However, this comparison has an important caveat: Mettack uses a GCN surrogate, so its poor transfer to IGNN reflects architectural mismatch as much as AEGIS's analytical superiority. The fair comparison is against the adaptive attacker (section V-D), which uses the same IFT information as AEGIS.

TABLE II: IFT vs. Mettack attack at $k = 1$ edge removal (10 seeds). IFT wins 30/30 comparisons. τ: Kendall correlation with brute-force ground truth.

Dataset	IFT τ	Met τ	IFT dmg	Met dmg
Cora	$+.28_{\pm.07}$	$+.20_{\pm.11}$	$.55_{\pm.27}$	$.16_{\pm.09}$
Citeseer	$+.38_{\pm.12}$	$+.16_{\pm.13}$	$1.1_{\pm1.5}$	$.11_{\pm.14}$
WikiCS	$+.14_{\pm.04}$	$+.52_{\pm.02}$	$.05_{\pm.01}$	$.01_{\pm.00}$

TABLE III: AEGIS deterministic vs. smoothing certificates (10 seeds). Smoothing at $\sigma \leq 0.10$ produces constant per-node radii (all nodes identical); AEGIS radii vary by node, reflecting structural vulnerability.

Dataset	AEGIS	Smoothing median radius			
		σ=.01	σ=.05	σ=.10	σ=.15
Cora	$.046_{\pm.010}$	.025	.123	$.244_{\pm.005}$	$.283_{\pm.132}$
Citeseer	$.045_{\pm.012}$	.025	.123	$.233_{\pm.013}$	$.091_{\pm.069}$
WikiCS	$.040_{\pm.004}$	.025	.123	$.246_{\pm.000}$	$.341_{\pm.017}$

Mettack's surrogate occasionally ranks edges closer to brute-force ground truth (WikiCS $\tau = +0.52$), but actual damage is $5\times$ lower because the GCN decision boundary diverges from the IGNN equilibrium.

C. Certificate Comparison: Deterministic vs. Smoothing

We compare AEGIS's deterministic first-order per-node radii (theorem 3) against randomized smoothing [3] with Gaussian noise on existing edges ($\sigma \in \{0.01, 0.05, 0.10, 0.15\}$, 1000 Monte Carlo samples, Clopper-Pearson confidence $1 - \alpha = 0.999$). table III reports median certified radius on Cora, Citeseer, and WikiCS (10 seeds, 50-node subgraphs).

The comparison reveals a qualitative difference rather than a simple radius ordering. At $\sigma \leq 0.10$, smoothing radii are *constant* across all nodes and datasets: every node receives the identical maximum-possible radius $r = \sigma \cdot \Phi^{-1}(p_{\text{lower}})$ because essentially all 1000 Monte Carlo samples predict correctly. This yields larger absolute radii than AEGIS (0.123 vs. 0.046 at $\sigma = 0.05$) but provides *zero per-node differentiation*—the certificates do not distinguish between vulnerable and robust nodes. At $\sigma = 0.15$, smoothing begins to differentiate (some nodes lose consistency), but with high variance.

AEGIS radii, by contrast, are *structurally informative*: median radius 0.041–0.046 with per-node variation reflecting actual equilibrium sensitivity. Nodes in dense, well-connected regions receive larger radii ($r_v \approx 0.09$), while boundary nodes get smaller ones ($r_v \approx 0.01$). This per-node differentiation is the key advantage for vulnerability ranking, not just worst-case bounds.

The two approaches are complementary: smoothing provides uniform probabilistic guarantees scaled by σ, while AEGIS provides tight, deterministic, structurally differentiated certificates that identify which specific nodes and edges are vulnerable.

TABLE IV: Adaptive attack (white-box PGD via IFT gradients on IGNN) vs. AEGIS first-order radii (10 seeds). Breach rate: fraction of nodes with $\varepsilon < r_v$ whose prediction changes under adaptive attack.

Dataset	ε	Breach	AEGIS dmg	Adapt. dmg	Ratio
Cora	0.01	0%	$0.33_{\pm0.07}$	$0.26_{\pm0.11}$	0.80
Cora	0.05	0%	$1.72_{\pm0.35}$	$1.32_{\pm0.55}$	0.76
Cora	0.10	0%	$3.70_{\pm0.79}$	$2.69_{\pm1.18}$	0.72
Citeseer	0.01	0%	$0.40_{\pm0.08}$	$0.34_{\pm0.06}$	0.84
Citeseer	0.05	0%	$2.14_{\pm0.42}$	$1.68_{\pm0.28}$	0.79
Citeseer	0.10	0%	$4.63_{\pm0.95}$	$3.37_{\pm0.55}$	0.74
WikiCS	0.01	0%	$0.21_{\pm0.02}$	$0.10_{\pm0.10}$	0.47
WikiCS	0.05	0.3%	$1.04_{\pm0.11}$	$0.48_{\pm0.49}$	0.46
WikiCS	0.10	0.6%	$2.10_{\pm0.22}$	$0.96_{\pm0.97}$	0.46

D. Adaptive Attack Evaluation

The Mettack comparison (section V-B) uses a GCN surrogate, so its failure against IGNN may reflect architectural mismatch rather than AEGIS's analytical strength. To rule this out, we implement an *adaptive attacker* that differentiates through the IGNN fixed-point iteration via implicit differentiation—the same IFT gradients AEGIS uses for sensitivity analysis—and optimizes perturbations directly against the IGNN loss surface using projected gradient descent (PGD, 50 steps, step size $\varepsilon/10$).

The adaptive attacker breaches zero nodes within their first-order radius at $\varepsilon = 0.01$ across all datasets and $< 1\%$ even at $\varepsilon = 0.10$, confirming empirical validity of the first-order radii in the small-budget regime. AEGIS's SVD-optimal attack consistently inflicts 20–50% more damage than adaptive PGD across all tested budgets (ratio 0.46–0.84). This is because the SVD direction is the globally optimal first-order perturbation, while PGD explores a non-convex landscape and can get trapped in local optima. The AEGIS advantage is largest on WikiCS (ratio 0.46), where the high-dimensional perturbation space makes PGD convergence hardest.

E. Phase Transition Validation

To validate theorem 1, we scale the adjacency matrix to achieve target spectral radii $\rho \in \{0.3, 0.5, 0.7, 0.85, 0.9, 0.95, 0.99\}$ and measure the actual fixed-point shift under a random perturbation of $\varepsilon = 0.01$. The shift amplifies by $83\times$ as ρ increases from 0.3 to 0.99, confirming the three-regime behavior: bounded shift in the subcritical regime, rapid growth near $\rho = 1$, and divergence in the supercritical regime.

F. Scalability

table V reports wall-clock time for the full AEGIS pipeline on Cora ego-subgraphs of varying size (single NVIDIA RTX 4090 GPU).

The practical GPU limit is $N \approx 200$ ($D = 12{,}800$). Empirical scaling is $16\times$ for $10\times$ growth in D, dominated by the $O(D \cdot N^2)$ numerical Jacobian J_A rather than the $O(D^3)$ linear solve. This covers all standard IEEE test cases (14–118 buses) and practical ego-network sizes for node-level analysis.

TABLE V: Scalability of AEGIS analysis on Cora ego-subgraphs.

N	$\lvert E\rvert$	D	J_z	J_A	Total
20	22	1,280	0.2s	0.2s	**0.5s**
50	61	3,200	0.5s	0.7s	**1.3s**
100	158	6,400	1.1s	1.6s	**2.8s**
200	389	12,800	2.3s	5.1s	**8.1s**

TABLE VI: First-order radius stability across subgraph sizes on Cora (10 seeds). Tightness and per-node radius are stable for $N \geq 50$.

N	Tight.	Med. r_v	Cert%	ρ	Time
30	$1.012_{\pm.003}$	$.064_{\pm.02}$	$89_{\pm 8}\%$	$.44_{\pm.04}$	0.5s
50	$1.013_{\pm.003}$	$.046_{\pm.01}$	$82_{\pm 8}\%$	$.45_{\pm.04}$	1.3s
100	$1.019_{\pm.004}$	$.030_{\pm.01}$	$84_{\pm 6}\%$	$.61_{\pm.05}$	10.3s
200	$1.031_{\pm.009}$	$.019_{\pm.01}$	$89_{\pm 3}\%$	$.75_{\pm.08}$	86.3s

G. Subgraph Size Ablation

The default $N = 50$ BFS ego-subgraph is critical to AEGIS's scalability. table VI evaluates whether certificate quality depends on subgraph size by varying $N \in \{30, 50, 100, 200\}$ on Cora (10 seeds).

Tightness is stable at 1.01–1.02 for $N \in \{30, 50, 100\}$ and degrades mildly to 1.031 ± 0.009 at $N = 200$, where the larger subgraph captures higher spectral radius ($\rho = 0.75$ vs. 0.45). Median radius *decreases* with N (from 0.064 at $N = 30$ to 0.019 at $N = 200$) because larger subgraphs include higher-degree nodes with smaller margins. Coverage remains stable at 82–89%. We recommend $N = 50$ as the default: tightness 1.013 ± 0.003 with 66× speedup over $N = 200$ (1.3s vs. 86s).

H. Hyperparameter Sensitivity

We evaluate sensitivity to the key hyperparameters: hidden dimension d and spectral-norm constraint c.

Hidden dimension. Varying $d \in \{16, 32, 64, 128\}$ on Cora: tightness is 1.012–1.013 for all values, confirming robustness to architecture width. Accuracy improves from 74.7% ($d = 16$) to 79.1% ($d = 64$). The critical budget $\varepsilon_{\text{crit}}$ and median radius are stable across d: $\varepsilon_{\text{crit}} \in [0.53, 0.56]$, $r_v \in [0.037, 0.051]$.

Spectral norm constraint. Varying $c \in \{0.5, 0.7, 0.9, 0.95\}$ reveals the accuracy–robustness tradeoff predicted by theorem 1. At $c = 0.5$: $\rho = 0.25$, $\varepsilon_{\text{crit}} = 1.50$, median $r_v = 0.090$—the largest certificates, but accuracy drops to 72.1%. At $c = 0.9$: $\rho = 0.45$, $\varepsilon_{\text{crit}} = 0.61$, $r_v = 0.051$, accuracy peaks at 80.6%. Radii shrink by 1.8× from $c = 0.5$ to $c = 0.9$ as $1/(1-\kappa)$ grows, consistent with the theoretical amplification factor.

I. DEQ Convergence Diagnostics

We report convergence statistics for the IGNN fixed-point iteration across all 9 datasets (10 seeds, relative tolerance $\|Z_{k+1} - Z_k\|_F < 10^{-5} \cdot \max(\|Z\|_F, 1)$). Convergence is 100% across all datasets and seeds. Graph benchmarks require 28 ± 7 (Amazon) to 113 ± 39 (Cora) iterations, while IEEE grids converge faster: 11 ± 2 (case14) to 44 ± 13 (case57). The non-normality index is $\eta = 1.02$–1.05 on graph benchmarks and $\eta = 1.20$–1.28 on IEEE grids, confirming mild non-normality across all domains. No seed on any dataset failed to converge, consistent with the enforced contractivity constraint $\kappa = \|J_z\|_2 < 1$ (via spectral normalization of W).

VI. Case Study: Power Flow Contingency

A natural question is whether AEGIS transfers beyond standard graph benchmarks. We demonstrate on a fundamentally different domain—AC power flow—where the adversarial vulnerability spectrum recovers a classical engineering problem: N-1 contingency analysis.

A. Background: N-1 Contingency

Power grid operators must ensure that no single transmission line failure causes system instability. The standard N-1 contingency assessment removes each line in turn, re-solves the power flow equations, and ranks lines by the severity of the resulting voltage/flow deviation. This brute-force procedure scales as $O(|E|)$ full power flow solves—expensive for large grids and infeasible for real-time operation.

We observe that AEGIS's per-edge vulnerability $v_{ij} = \|S_{:,iN+j} + S_{:,jN+i}\|$ computes an analogous quantity via a single IFT analysis. The mathematical correspondence is approximate (continuous first-order sensitivity vs. discrete full-removal contingency):

ML concept	**Power systems concept**
Edge weight perturbation δA_{ij}	Topology change (line trip)
Vulnerability spectrum v_{ij}	Contingency severity ranking
Critical budget $\varepsilon_{\text{crit}}$	Stability margin
Contractivity loss $\kappa \to 1$	Approach to voltage collapse

Scope of the analogy. This correspondence is approximate: physical line outages are discrete (full edge removal), whereas AEGIS analyzes continuous edge-weight perturbations. Line impedance is fixed; what changes under contingency is the topology (which lines are in service) and the resulting power flow redistribution. AEGIS captures this as a first-order sensitivity to adjacency changes, which is analogous to how Power Transfer Distribution Factors (PTDF) and Line Outage Distribution Factors (LODF) [11] linearize the effect of generation shifts and line trips, respectively.

B. Experimental Setup

We train ContractiveGCN-PF—an IGNN variant for AC power flow—on four standard IEEE test cases. Training data is generated via PandaPower's full AC Newton-Raphson solver (2,000 load samples per case, uniformly sampled at 70–130% of nominal load). The model takes binary adjacency from the admittance matrix and 5 bus features: bus type indicators (slack, PV), active/reactive power injections (P, Q), and voltage magnitude $|V|$. Unlike graph benchmarks, power flow grids are small enough (14–118 buses) to analyze without subgraph extraction.

Limitation: sample diversity. The 2,000 samples per case cover uniform load scaling but not seasonal variation, renewable intermittency, or generator outage combinations. This limits the generalizability of the learned GNN model; the AEGIS analysis itself (IFT-based sensitivity) is exact given any trained model, but the model's fidelity to real operating conditions depends on training data diversity. Scaling to larger sample sets with realistic operating profiles is straightforward but requires representative scenario generation.

C. Results

table VII reports constrained tightness and N-1 ranking correlation (Kendall τ between AEGIS vulnerability spectrum and brute-force contingency ranking) across 10 seeds.

Key observations. (1) Tightness of ≈ 1.00 under constrained perturbations transfers to the power flow domain, confirming that the first-order prediction property is not domain-specific. (2) Ranking correlation (τ) is moderate (0.37–0.67), reflecting the gap between continuous first-order sensitivity and discrete full-edge-removal contingency. The operationally relevant metric is *precision at top-k*: AEGIS achieves P@5 = 0.52–0.70 and P@10 = 0.66–0.81, meaning it correctly identifies 3–4 of the 5 most critical lines and 7–8 of the top 10. P@10 on case118 (0.81) is particularly strong. (3) We compare against an effective-resistance baseline derived from the graph Laplacian pseudoinverse (a simplified DC-approximation proxy for LODF [11]). On small grids (case14/30), this baseline achieves $\tau \approx 0.37$, comparable to AEGIS. On larger grids (case57/118), the simplified baseline degrades ($\tau \leq 0.04$) while AEGIS maintains $\tau = 0.62$–0.67. We note that the simplified baseline uses unit-impedance (binary adjacency), not actual line reactances; a proper LODF implementation with real impedance data would likely perform better, but requires domain-specific information unavailable to AEGIS.

Practical positioning. AEGIS is not a replacement for PTDF/LODF screening, which exploits domain-specific DC power flow linearization with real line parameters. Rather, it demonstrates that general-purpose IFT-based vulnerability analysis *independently recovers* contingency structure without domain-specific inputs. For case118, AEGIS computes the ranking in a single IFT analysis (2.3s) versus 179 full AC solves for brute-force N-1, with P@10 = 0.81.

Limitation: operational readiness. The $\tau = 0.37$–0.67 correlation is insufficient for direct operational use without verification. Grid operators require near-perfect recall of critical contingencies. AEGIS is positioned as a *fast screening layer*, not a standalone contingency tool.

VII. Related Work

Adversarial attacks on GNNs. Structural attacks modify graph topology to degrade GNN performance. Nettack [1] targets individual nodes via greedy edge flips; Mettack [2] uses meta-gradients through a GCN surrogate for global poisoning. These methods operate on explicit (finite-depth) GNNs and require surrogate transfer when applied to DEQ models, which we show causes significant performance degradation due to architectural mismatch (sections V-B and V-D). AEGIS directly analyzes the equilibrium via IFT, avoiding surrogate transfer entirely.

Certified robustness for GNNs. Bojchevski and Günnemann [12] provide the first certifiable robustness analysis for GNNs under structural perturbation, using convex relaxations for GCN-class models; their approach requires finite-depth explicit propagation and does not apply to implicit fixed-point models. Randomized smoothing [3] extends to probabilistic certificates by classifying under random graph perturbation; it provides valid certificates at arbitrary perturbation magnitudes but degrades base accuracy with increasing noise σ. Li et al. [13] propose AGNNCert with deterministic guarantees for GNNs under arbitrary perturbations, but again target explicit architectures. Interval bound propagation [14] requires layer-by-layer propagation, incompatible with infinite-depth DEQ models. AEGIS provides deterministic first-order robustness radii for the implicit (IGNN-class) setting; these are structurally differentiated per-node, unlike smoothing's uniform bounds per noise level (section V-C).

DEQ robustness and Lipschitz analysis. Winston and Kolter [15] develop monotone operator equilibrium networks with built-in Lipschitz bounds, providing robustness certificates for input perturbations by construction. Revay et al. [10] bound the Lipschitz constant of equilibrium networks via linear matrix inequalities. Pabbaraju et al. [16] estimate tighter Lipschitz constants for monotone DEQs, directly bounding $\|\partial z^*/\partial x\|$. El Ghaoui et al. [9] analyze implicit deep learning with a focus on well-posedness and input sensitivity. All of these works address *input* (feature) perturbation sensitivity $\|\partial z^*/\partial x\|$, not *structural* sensitivity $\|\partial z^*/\partial A\|$. The key distinction is that structural perturbation changes the operator itself (via A), not just its input—the fixed point depends on A through both J_z and J_A. AEGIS addresses this gap with the constrained structural sensitivity matrix S_c.

Implicit GNNs. IGNN [4] enforces contractivity via spectral normalization of W, guaranteeing unique equilibria. EIGNN [17] improves computational efficiency via eigendecomposition. Neither analyzes adversarial structural vulnerability at the equilibrium. Liu et al. [18] study stability of GNNs under graph perturbation but without certified per-node radii or the explicit three-regime characterization.

Scope and limitations. AEGIS's theoretical analysis applies specifically to contractive implicit GNNs (IGNN-class models) that factorize as $F(Z, A) = \sigma(AZW^\top + b)$. Most deployed GNNs (GAT, GIN, GraphSAGE) do not have this well-separated A/W structure and lack guaranteed fixed points, so AEGIS cannot be directly applied. Extension to other implicit architectures (monotone operator networks [15], neural ODEs on graphs) is a promising direction but requires architecture-specific IFT derivations.

TABLE VII: AEGIS on IEEE power flow benchmarks (10 seeds). Tightness ≈ 1.00 under constrained perturbations. P@k: precision of AEGIS top-k vs. brute-force N-1 ground truth. EffRes: effective-resistance baseline (simplified LODF proxy).

				AEGIS	AEGIS Precision		EffRes Baseline	
Case	N	$\|E\|$	Tight.	τ	P@5	P@10	τ	P@10
case14	14	20	$1.00_{\pm.01}$	$+.42_{\pm.19}$	$.70_{\pm.10}$	$.74_{\pm.12}$	$+.37_{\pm.09}$	$.60_{\pm.06}$
case30	30	41	$1.00_{\pm.01}$	$+.37_{\pm.17}$	$.66_{\pm.13}$	$.68_{\pm.06}$	$+.37_{\pm.05}$	$.40_{\pm.08}$
case57	57	78	$1.02_{\pm.01}$	$+.67_{\pm.09}$	$.52_{\pm.24}$	$.66_{\pm.13}$	$-.14_{\pm.11}$	$.06_{\pm.05}$
case118	118	179	$1.01_{\pm.01}$	$+.62_{\pm.11}$	$.62_{\pm.14}$	$.81_{\pm.10}$	$+.04_{\pm.08}$	$.10_{\pm.00}$

VIII. Conclusion

We presented AEGIS, a framework that exploits the equilibrium structure of contractive implicit GNNs (IGNN-class) for adversarial vulnerability analysis. Using the operator-norm contraction constant $\kappa = \|J_z\|_2$ and the IFT, we formalized vulnerability regimes around a critical budget $\varepsilon_{\text{crit}} = (1-\kappa)/\|W\|_2$, with explicit assumptions (ReLU, spectral-norm constraint, linear head). The central technical contribution—the constrained sensitivity matrix S_c—achieves first-order tightness 1.00 ± 0.01 at $\varepsilon = 0.01$ across 9 datasets. AEGIS's SVD-optimal attack inflicts 20–50% more damage than adaptive white-box PGD. Its deterministic first-order per-node robustness radii provide structurally differentiated vulnerability assessments, unlike randomized smoothing's uniform bounds. The vulnerability spectrum independently recovers power grid N-1 contingency rankings (P@10 = 0.66–0.81), bridging adversarial ML with power systems analysis.

Limitations. (1) *Architecture scope:* AEGIS applies to contractive implicit GNNs (IGNN-class) only, not to GAT, GIN, or other explicit architectures. (2) *First-order approximation:* radii are locally tight for small ε but conservative at large budgets ($\varepsilon > 0.1$); without bounding the second-order remainder, they are local sensitivity radii, not global robustness certificates. Edge additions that change sparsity pattern break the continuous perturbation assumption entirely. (3) *Threat model:* we perturb normalized adjacency $\hat{A}$ directly; perturbations to raw A that change degree normalization require additional chain-rule terms not currently included. (4) *Scalability:* dense Jacobian computation limits subgraph size to $N \approx 200$. (5) *Power flow:* $\tau = 0.37$–0.67 is insufficient for standalone operational use; AEGIS serves as a screening layer, not a replacement for PTDF/LODF analysis with real line parameters.

Future work. Sparse Neumann-series solvers for larger-scale analysis. Extension to monotone operator networks [15] and neural ODEs on graphs. Discrete perturbation models (edge insertion/deletion) via combinatorial optimization over the constrained sensitivity spectrum.

References

[1] D. Zügner, A. Akbarnejad, and S. Günnemann, "Adversarial attacks on neural networks for graph data," in *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2018, pp. 2847–2856.

[2] D. Zügner and S. Günnemann, "Adversarial attacks on graph neural networks via meta learning," in *International Conference on Learning Representations*, 2019.

[3] A. Bojchevski, J. Klicpera, and S. Günnemann, "Efficient robustness certificates for discrete data: Sparsity-aware randomized smoothing for graphs, images and more," in *International Conference on Machine Learning*, 2020, pp. 1003–1013.

[4] F. Gu, H. Chang, W. Zhu, S. Sojoudi, and L. El Ghaoui, "Implicit graph neural networks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 11 984–11 995.

[5] S. Bai, J. Z. Kolter, and V. Koltun, "Deep equilibrium models," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[6] L. N. Trefethen and M. Embree, *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*. Princeton University Press, 2005.

[7] J. Lorraine, P. Vicol, and D. Duvenaud, "Optimizing millions of hyperparameters by implicit differentiation," in *International Conference on Artificial Intelligence and Statistics*, 2020, pp. 1540–1552.

[8] S. Gould, R. Hartley, and D. Campbell, "Deep declarative networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 8, pp. 3988–4004, 2021.

[9] L. El Ghaoui, F. Gu, B. Travacca, A. Askari, and A. Tsai, "Implicit deep learning," *SIAM Journal on Mathematics of Data Science*, vol. 3, no. 3, pp. 930–958, 2021.

[10] M. Revay, R. Wang, and I. R. Manchester, "Lipschitz bounded equilibrium networks," in *arXiv preprint arXiv:2010.01732*, 2020.

[11] A. J. Wood, B. F. Wollenberg, and G. B. Sheblé, *Power Generation, Operation, and Control*, 3rd ed. John Wiley & Sons, 2014.

[12] A. Bojchevski and S. Günnemann, "Certifiable robustness to graph perturbations," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[13] J. Li, G. Tao, D. Han, and X. Zhang, "AGNNCert: Defending graph neural networks against arbitrary perturbations with certified robustness," in *USENIX Security Symposium*, 2025.

[14] D. Zügner and S. Günnemann, "Certifiable robustness and robust training for graph convolutional networks," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2019, pp. 246–256.

[15] E. Winston and J. Z. Kolter, "Monotone operator equilibrium networks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 10 718–10 728.

[16] C. Pabbaraju, E. Winston, and J. Z. Kolter, "Estimating lipschitz constants of monotone deep equilibrium models," *arXiv preprint arXiv:2111.08931*, 2021.

[17] J. Liu, K. Kawaguchi, B. Hooi, Y. Wang, and J. Feng, "Eignn: Efficient infinite-depth graph neural networks," in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 18 762–18 773.

[18] S. Liu, R. Ying, H. Dong, L. Li, T. Xu, Y. Liang, J. Leskovec, and H. Peng, "Stability and generalization of graph neural networks," in *arXiv preprint*, 2021.